

Experiment No. 8

Student Name: Ojaswini Thote

PRN: 24070521048

Experiment Title

Implementation of Form Validation and Event Handling using JavaScript; Development of a Gym Membership Registration Form

Software / Tools Required

- Visual Studio Code
- Google Chrome
- HTML5
- JavaScript (ES6)

Practical — FitSpace Gym Membership Registration

Aim

To develop a responsive gym membership registration form using HTML, CSS, and JavaScript, with client-side form validation and event handling.

File Path

<file:///Users/ojaswinithote/Desktop/Javascript/EXP%208/Practical/index.html>

file:///Users/ojaswinithote/Desktop/Javascript/EXP%208/Case%20Study/index2.html

Practical/script.js

Program Code

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>FitSpace Gym Membership</title>

  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&family=Oswald:wght@500;600;700&display=swap"
rel="stylesheet">

  <style>
    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
```

```

    font-family: "DM Sans", sans-serif;
    background: #f3f0e8;
    color: #28322b;
    min-height: 100vh;
    padding: 35px 20px;
}

.container {
    max-width: 850px;
    margin: auto;
    background: #ffffdf8;
    border-radius: 20px;
    overflow: hidden;
    box-shadow: 0 12px 35px rgba(48, 58, 49, 0.12);
}

.header {
    background: #b8c7ad;
    padding: 35px 40px;
    position: relative;
    overflow: hidden;
}

.header::after {
    content: "FIT";
    position: absolute;
    right: 25px;
    bottom: -20px;
    font-family: "Oswald", sans-serif;
    font-size: 110px;
    font-weight: 700;
    color: rgba(255, 255, 255, 0.25);
}

.header h1 {
    font-family: "Oswald", sans-serif;
    font-size: 42px;
    text-transform: uppercase;
    letter-spacing: 1px;
    color: #26332a;
    position: relative;
    z-index: 1;
}

.header p {
    margin-top: 7px;
    font-size: 15px;
    color: #465448;
    position: relative;
    z-index: 1;
}

form {
    padding: 35px 40px 40px;
}

.section-title {
    font-family: "Oswald", sans-serif;
    text-transform: uppercase;
    font-size: 21px;
    letter-spacing: 0.8px;
    margin-bottom: 22px;
    color: #52634f;
}

.form-grid {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 20px;

```

```

}

.form-group {
  display: flex;
  flex-direction: column;
}

.full-width {
  grid-column: 1 / -1;
}

label {
  font-weight: 600;
  font-size: 14px;
  margin-bottom: 7px;
  color: #39463c;
}

input,
select {
  width: 100%;
  padding: 13px 14px;
  border: 2px solid #d7ddd2;
  border-radius: 9px;
  background: #fbfaf6;
  font-family: "DM Sans", sans-serif;
  font-size: 14px;
  color: #303a32;
  outline: none;
  transition: 0.2s ease;
}

input:focus,
select:focus {
  border-color: #7e9674;
  background: #f5f8f1;
}

input.valid,
select.valid {
  border-color: #7e9674;
  background: #f2f7ef;
}

input.invalid,
select.invalid {
  border-color: #c87878;
  background: #fff5f5;
}

.error {
  color: #bd5d5d;
  font-size: 12px;
  margin-top: 5px;
  min-height: 16px;
}

.gender-options {
  display: flex;
  gap: 18px;
  padding: 11px 0;
}

.gender-options label {
  display: flex;
  align-items: center;
  gap: 6px;
  font-weight: 500;
  margin: 0;
}

```

```

        cursor: pointer;
    }

    .gender-options input {
        width: auto;
        accent-color: #718768;
    }

    .checkbox-group {
        display: flex;
        align-items: center;
        gap: 9px;
        margin-top: 8px;
    }

    .checkbox-group input {
        width: auto;
        accent-color: #718768;
    }

    .checkbox-group label {
        margin: 0;
        font-weight: 500;
        cursor: pointer;
    }

    button {
        width: 100%;
        margin-top: 25px;
        padding: 15px;
        border: none;
        border-radius: 10px;
        background: #64795d;
        color: white;
        font-family: "Oswald", sans-serif;
        font-size: 18px;
        text-transform: uppercase;
        letter-spacing: 1px;
        cursor: pointer;
        transition: 0.2s ease;
    }

    button:hover {
        background: #52674d;
        transform: translateY(-1px);
    }

    #message {
        text-align: center;
        margin-top: 18px;
        padding: 12px;
        border-radius: 8px;
        font-size: 14px;
        font-weight: 600;
        display: none;
    }

    #message.success {
        display: block;
        background: #e4efdf;
        color: #4d6848;
    }

    #message.fail {
        display: block;
        background: #f9e5e5;
        color: #a05252;
    }

```

```

@media (max-width: 650px) {
  .form-grid {
    grid-template-columns: 1fr;
  }

  .full-width {
    grid-column: auto;
  }

  .header,
  form {
    padding: 28px 22px;
  }

  .header h1 {
    font-size: 34px;
  }
}
</style>
</head>

<body>

<div class="container">

  <div class="header">
    <h1>FitSpace Gym</h1>
    <p>Build strength. Stay consistent. Become your best.</p>
  </div>

  <form id="membershipForm">

    <h2 class="section-title">Membership Registration</h2>

    <div class="form-grid">

      <div class="form-group full-width">
        <label for="fullName">Full Name</label>
        <input type="text" id="fullName" placeholder="Enter your full name">
        <span class="error" id="nameError"></span>
      </div>

      <div class="form-group">
        <label for="age">Age</label>
        <input type="number" id="age" min="16" max="60" placeholder="16 - 60">
        <span class="error" id="ageError"></span>
      </div>

      <div class="form-group">
        <label>Gender</label>

        <div class="gender-options">
          <label>
            <input type="radio" name="gender" value="Male">
            Male
          </label>

          <label>
            <input type="radio" name="gender" value="Female">
            Female
          </label>

          <label>
            <input type="radio" name="gender" value="Other">
            Other
          </label>
        </div>

        <span class="error" id="genderError"></span>
      </div>
    </div>
  </form>
</div>

```

```

</div>

<div class="form-group">
  <label for="email">Email Address</label>
  <input type="email" id="email" placeholder="example@email.com">
  <span class="error" id="emailError"></span>
</div>

<div class="form-group">
  <label for="phone">Phone Number</label>
  <input type="text" id="phone" maxlength="10" placeholder="10 digit number">
  <span class="error" id="phoneError"></span>
</div>

<div class="form-group">
  <label for="goal">Fitness Goal</label>

  <select id="goal">
    <option value="">Select your fitness goal</option>
    <option value="Weight Loss">Weight Loss</option>
    <option value="Muscle Gain">Muscle Gain</option>
    <option value="General Fitness">General Fitness</option>
    <option value="Strength Training">Strength Training</option>
  </select>

  <span class="error" id="goalError"></span>
</div>

<div class="form-group">
  <label for="plan">Membership Plan</label>

  <select id="plan">
    <option value="">Select a membership plan</option>
    <option value="Monthly">Monthly – Flexible</option>
    <option value="Quarterly">Quarterly – 3 Months</option>
    <option value="Yearly">Yearly – Best Value</option>
  </select>

  <span class="error" id="planError"></span>
</div>

<div class="form-group full-width">

  <div class="checkbox-group">
    <input type="checkbox" id="terms">

    <label for="terms">
      I agree to the gym membership terms and conditions.
    </label>
  </div>

  <span class="error" id="termsError"></span>

</div>

</div>

<button type="submit">Register Membership</button>

<div id="message"></div>

</form>

</div>

<script src="script.js"></script>

</body>
</html>

```

script.js

```
const form = document.getElementById("membershipForm");

const fullName = document.getElementById("fullName");
const age = document.getElementById("age");
const email = document.getElementById("email");
const phone = document.getElementById("phone");
const goal = document.getElementById("goal");
const plan = document.getElementById("plan");
const terms = document.getElementById("terms");

const nameError = document.getElementById("nameError");
const ageError = document.getElementById("ageError");
const emailError = document.getElementById("emailError");
const phoneError = document.getElementById("phoneError");
const goalError = document.getElementById("goalError");
const planError = document.getElementById("planError");
const termsError = document.getElementById("termsError");

const message = document.getElementById("message");

function validateName() {
    const value = fullName.value.trim();

    if (value === "") {
        nameError.textContent = "Full name is required.";
        fullName.classList.add("invalid");
        fullName.classList.remove("valid");
        return false;
    }

    if (!/^[A-Za-z ]+$/ .test(value)) {
        nameError.textContent = "Name should contain only letters.";
        fullName.classList.add("invalid");
        fullName.classList.remove("valid");
        return false;
    }

    nameError.textContent = "";
    fullName.classList.add("valid");
    fullName.classList.remove("invalid");
    return true;
}

function validateAge() {
    const value = Number(age.value);

    if (age.value === "") {
        ageError.textContent = "Age is required.";
        age.classList.add("invalid");
        age.classList.remove("valid");
        return false;
    }

    if (value < 16 || value > 60) {
        ageError.textContent = "Age must be between 16 and 60.";
        age.classList.add("invalid");
        age.classList.remove("valid");
        return false;
    }

    ageError.textContent = "";
    age.classList.add("valid");
    age.classList.remove("invalid");
    return true;
}

function validateGender() {
```

```

const gender = document.querySelector('input[name="gender"]:checked');

if (!gender) {
    genderError.textContent = "Please select your gender.";
    return false;
}

genderError.textContent = "";
return true;
}

function validateEmail() {
    const value = email.value.trim();
    const pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

    if (value === "") {
        emailError.textContent = "Email is required.";
        email.classList.add("invalid");
        email.classList.remove("valid");
        return false;
    }

    if (!pattern.test(value)) {
        emailError.textContent = "Enter a valid email address.";
        email.classList.add("invalid");
        email.classList.remove("valid");
        return false;
    }

    emailError.textContent = "";
    email.classList.add("valid");
    email.classList.remove("invalid");
    return true;
}

function validatePhone() {
    const value = phone.value.trim();

    if (value === "") {
        phoneError.textContent = "Phone number is required.";
        phone.classList.add("invalid");
        phone.classList.remove("valid");
        return false;
    }

    if (!/^[0-9]{10}$/.test(value)) {
        phoneError.textContent = "Phone number must contain 10 digits.";
        phone.classList.add("invalid");
        phone.classList.remove("valid");
        return false;
    }

    phoneError.textContent = "";
    phone.classList.add("valid");
    phone.classList.remove("invalid");
    return true;
}

function validateGoal() {
    if (goal.value === "") {
        goalError.textContent = "Please select a fitness goal.";
        goal.classList.add("invalid");
        goal.classList.remove("valid");
        return false;
    }

    goalError.textContent = "";
    goal.classList.add("valid");
    goal.classList.remove("invalid");

```

```

        return true;
    }

    function validatePlan() {
        if (plan.value === "") {
            planError.textContent = "Please select a membership plan.";
            plan.classList.add("invalid");
            plan.classList.remove("valid");
            return false;
        }

        planError.textContent = "";
        plan.classList.add("valid");
        plan.classList.remove("invalid");
        return true;
    }

    function validateTerms() {
        if (!terms.checked) {
            termsError.textContent = "You must agree to the terms and conditions.";
            return false;
        }

        termsError.textContent = "";
        return true;
    }

    fullName.addEventListener("focus", function () {
        fullName.style.backgroundColor = "#f5f8f1";
    });

    fullName.addEventListener("input", validateName);

    fullName.addEventListener("blur", function () {
        validateName();
        fullName.style.backgroundColor = "";
    });

    age.addEventListener("focus", function () {
        age.style.backgroundColor = "#f5f8f1";
    });

    age.addEventListener("input", validateAge);

    age.addEventListener("blur", function () {
        validateAge();
        age.style.backgroundColor = "";
    });

    document.querySelectorAll('input[name="gender"]').forEach(function (radio) {
        radio.addEventListener("change", validateGender);
    });

    email.addEventListener("focus", function () {
        email.style.backgroundColor = "#f5f8f1";
    });

    email.addEventListener("input", validateEmail);

    email.addEventListener("blur", function () {
        validateEmail();
        email.style.backgroundColor = "";
    });

    phone.addEventListener("focus", function () {
        phone.style.backgroundColor = "#f5f8f1";
    });

    phone.addEventListener("input", function () {

```

```

        phone.value = phone.value.replace(/\D/g, "");
        validatePhone();
    });

    phone.addEventListener("blur", function () {
        validatePhone();
        phone.style.backgroundColor = "";
    });

    goal.addEventListener("focus", function () {
        goal.style.backgroundColor = "#f5f8f1";
    });

    goal.addEventListener("change", validateGoal);

    goal.addEventListener("blur", function () {
        validateGoal();
        goal.style.backgroundColor = "";
    });

    plan.addEventListener("focus", function () {
        plan.style.backgroundColor = "#f5f8f1";
    });

    plan.addEventListener("change", validatePlan);

    plan.addEventListener("blur", function () {
        validatePlan();
        plan.style.backgroundColor = "";
    });

    terms.addEventListener("change", validateTerms);

    form.addEventListener("submit", function (event) {
        event.preventDefault();

        const validName = validateName();
        const validAge = validateAge();
        const validGender = validateGender();
        const validEmail = validateEmail();
        const validPhone = validatePhone();
        const validGoal = validateGoal();
        const validPlan = validatePlan();
        const validTerms = validateTerms();

        if (
            validName &&
            validAge &&
            validGender &&
            validEmail &&
            validPhone &&
            validGoal &&
            validPlan &&
            validTerms
        ) {
            message.textContent = "Membership successfully registered!";
            message.className = "success";

            form.reset();

            document.querySelectorAll("input, select").forEach(function (field) {
                field.classList.remove("valid");
            });
        } else {
            message.textContent = "Please correct the highlighted fields.";
            message.className = "fail";
        }
    });

```

Form Fields

Field	Purpose / Validation
Full Name	Accepts the member's full name.
Age	Accepts an age within the specified 16–60 range.
Gender	Allows selection of Male, Female, or Other.
Email Address	Accepts and validates the member's email address.
Phone Number	Accepts a 10-digit phone number.
Fitness Goal	Allows selection of a fitness goal.
Membership Plan	Allows selection of Monthly, Quarterly, or Yearly plan.
Terms and Conditions	Requires the user to agree before registration.

JavaScript Concepts / Functions Used

Concept / Function	Purpose
DOM Selection	Accesses form elements for validation and interaction.
addEventListener()	Handles form submission and input-related events.
Form Validation	Checks required fields and validates user input.
Regular Expressions	Validates formats such as email and phone number.
preventDefault()	Prevents invalid form submission.
classList	Applies valid or invalid styles to fields.
textContent	Displays validation messages.

Output

- The FitSpace Gym membership form is displayed with a responsive layout.
- Users can enter personal information and select membership details.
- Validation provides appropriate error messages for invalid or missing information.
- Valid and invalid fields are visually distinguished.
- The terms and conditions checkbox is required.
- A success message is displayed after successful validation.

Screenshot

FITSPACE GYM
Build strength. Stay consistent. Become your best.

MEMBERSHIP REGISTRATION

Full Name
Ojaswini Thote

Age
20

Gender
☐ Male ☒ Female ☐ Other

Email Address
ojaswinithote@gmail.com

Phone Number
8767064228

Fitness Goal
Strength Training

Membership Plan
Quarterly - 3 Months

☒ I agree to the gym membership terms and conditions.

REGISTER MEMBERSHIP

Figure 1: FitSpace Gym Membership Registration — Practical Output

Case Study — FitSpace Gym Membership Registration with Additional Information

Aim

To enhance the gym membership registration form by collecting additional member information such as eye color, physical characteristics, and athletic ability, while applying JavaScript validation and event handling.

File Path

Case Study/index2.html

Case Study/script2.js

Program Code

index2.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>FitSpace Gym Membership</title>
```

```
<style>
```

```
    * {
        box-sizing: border-box;
        margin: 0;
        padding: 0;
    }

    body {
        font-family: Arial, Helvetica, sans-serif;
        background: #f3f0e8;
        color: #28322b;
        min-height: 100vh;
        padding: 35px 20px;
    }

    .container {
        max-width: 850px;
        margin: auto;
        background: #fffdf8;
        border-radius: 20px;
        overflow: hidden;
        box-shadow: 0 12px 35px rgba(48, 58, 49, 0.12);
    }

    /* ===== HEADER ===== */

    .header {
        background: #b8c7ad;
        padding: 35px 40px;
        position: relative;
        overflow: hidden;
    }

    .header::after {
        content: "FIT";
        position: absolute;
        right: 25px;
        bottom: -20px;
        font-size: 110px;
        font-weight: 700;
        color: rgba(255, 255, 255, 0.25);
    }

    .header h1 {
        font-size: 42px;
        text-transform: uppercase;
        letter-spacing: 1px;
        color: #26332a;
        position: relative;
        z-index: 1;
    }

    .header h1 span {
        color: #52634f;
    }

    .header p {
        margin-top: 7px;
        font-size: 15px;
        color: #465448;
        position: relative;
        z-index: 1;
    }
```

```

/* ===== FORM ===== */

form {
  padding: 35px 40px 40px;
}

.section-title {
  text-transform: uppercase;
  font-size: 21px;
  letter-spacing: 0.8px;
  margin-bottom: 22px;
  color: #52634f;
}

.form-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 20px;
}

.form-group {
  display: flex;
  flex-direction: column;
}

.full-width {
  grid-column: 1 / -1;
}

label {
  font-weight: 600;
  font-size: 14px;
  margin-bottom: 7px;
  color: #39463c;
}

input,
select,
textarea {
  width: 100%;
  padding: 13px 14px;
  border: 2px solid #d7ddd2;
  border-radius: 9px;
  background: #fbfaf6;
  font-family: Arial, Helvetica, sans-serif;
  font-size: 14px;
  color: #303a32;
  outline: none;
  transition: 0.2s ease;
}

input:focus,
select:focus,
textarea:focus {
  border-color: #7e9674;
  background: #f5f8f1;
}

input.valid,
select.valid,
textarea.valid {
  border-color: #7e9674;
  background: #f2f7ef;
}

input.invalid,
select.invalid,
textarea.invalid {
  border-color: #c87878;
}

```

```

        background: #fff5f5;
    }

    .error {
        color: #bd5d5d;
        font-size: 12px;
        margin-top: 5px;
        min-height: 16px;
    }

/* ===== GENDER ===== */

.gender-options {
    display: flex;
    gap: 18px;
    padding: 11px 0;
}

.gender-options label {
    display: flex;
    align-items: center;
    gap: 6px;
    font-weight: 500;
    margin: 0;
    cursor: pointer;
}

.gender-options input {
    width: auto;
    accent-color: #718768;
}

/* ===== CHECKBOXES ===== */

.checkbox-options {
    display: flex;
    flex-direction: column;
    gap: 11px;
    padding: 5px 0;
}

.checkbox-option {
    display: flex;
    align-items: center;
    gap: 9px;
    font-weight: 500;
    cursor: pointer;
}

.checkbox-option input {
    width: auto;
    accent-color: #718768;
}

/* ===== TEXTAREA ===== */

textarea {
    min-height: 120px;
    resize: vertical;
}

/* ===== BUTTON ===== */

button {
    width: 100%;
    margin-top: 25px;
    padding: 15px;
    border: none;
    border-radius: 10px;

```

```

        background: #64795d;
        color: white;
        font-size: 17px;
        font-weight: bold;
        text-transform: uppercase;
        letter-spacing: 1px;
        cursor: pointer;
        transition: 0.2s ease;
    }

    button:hover {
        background: #52674d;
        transform: translateY(-1px);
    }

    /* ===== MESSAGE ===== */

```

```

#message {
    text-align: center;
    margin-top: 18px;
    padding: 12px;
    border-radius: 8px;
    font-size: 14px;
    font-weight: 600;
    display: none;
}

```

```

#message.success {
    display: block;
    background: #e4efdf;
    color: #4d6848;
}

```

```

#message.fail {
    display: block;
    background: #f9e5e5;
    color: #a05252;
}

```

```

/* ===== RESPONSIVE ===== */

```

```

@media (max-width: 650px) {

    .form-grid {
        grid-template-columns: 1fr;
    }

    .full-width {
        grid-column: auto;
    }

    .header,
    form {
        padding: 28px 22px;
    }

    .header h1 {
        font-size: 34px;
    }

    .gender-options {
        flex-direction: column;
        gap: 12px;
    }
}

```

</style>

</head>

```
<body>

<div class="container">

  <!-- ===== HEADER ===== -->

  <div class="header">

    <h1>
      Fit<span>Space</span> Gym
    </h1>

    <p>
      Build strength. Stay consistent. Become your best.
    </p>

  </div>

  <!-- ===== FORM ===== -->

  <form id="membershipForm">

    <h2 class="section-title">
      Membership Registration
    </h2>

    <div class="form-grid">

      <!-- FULL NAME -->

      <div class="form-group full-width">

        <label for="fullName">
          Full Name
        </label>

        <input
          type="text"
          id="fullName"
          placeholder="Enter your full name"
        >

        <span
          class="error"
          id="nameError"
        ></span>

      </div>

      <!-- AGE -->

      <div class="form-group">

        <label for="age">
          Age
        </label>

        <input
          type="number"
          id="age"
        >

      </div>

    </div>

  </form>

</div>
```

```

        min="16"
        max="60"
        placeholder="16 - 60"
    >

    <span
        class="error"
        id="ageError"
    ></span>

</div>

<!-- GENDER -->

<div class="form-group">

    <label>
        Gender
    </label>

    <div class="gender-options">

        <label>

            <input
                type="radio"
                name="gender"
                value="Male"
            >

            Male

        </label>

        <label>

            <input
                type="radio"
                name="gender"
                value="Female"
            >

            Female

        </label>

        <label>

            <input
                type="radio"
                name="gender"
                value="Other"
            >

            Other

        </label>

    </div>

    <span
        class="error"
        id="genderError"
    ></span>

</div>

```

```
<!-- EMAIL -->

<div class="form-group">

    <label for="email">
        Email Address
    </label>

    <input
        type="email"
        id="email"
        placeholder="example@email.com"
    >

    <span
        class="error"
        id="emailError"
    ></span>

</div>
```

```
<!-- PHONE -->

<div class="form-group">

    <label for="phone">
        Phone Number
    </label>

    <input
        type="text"
        id="phone"
        maxlength="10"
        placeholder="10 digit number"
    >

    <span
        class="error"
        id="phoneError"
    ></span>

</div>
```

```
<!-- EYE COLOR -->

<div class="form-group">

    <label for="eyeColor">
        Eye Color
    </label>

    <select id="eyeColor">

        <option value="">
            Select eye color
        </option>

        <option value="Green">
            Green
        </option>

        <option value="Blue">
            Blue
        </option>

    </select>

</div>
```

```

        <option value="Brown">
            Brown
        </option>

        <option value="Hazel">
            Hazel
        </option>

        <option value="Gray">
            Gray
        </option>

    </select>

    <span
        class="error"
        id="eyeError"
    ></span>

</div>

<!-- FITNESS GOAL -->

<div class="form-group">

    <label for="goal">
        Fitness Goal
    </label>

    <select id="goal">

        <option value="">
            Select your fitness goal
        </option>

        <option value="Weight Loss">
            Weight Loss
        </option>

        <option value="Muscle Gain">
            Muscle Gain
        </option>

        <option value="General Fitness">
            General Fitness
        </option>

        <option value="Strength Training">
            Strength Training
        </option>

    </select>

    <span
        class="error"
        id="goalError"
    ></span>

</div>

<!-- MEMBERSHIP PLAN -->

<div class="form-group">

    <label for="plan">
        Membership Plan
    </label>

```

```

<select id="plan">

    <option value="">
        Select a membership plan
    </option>

    <option value="Monthly">
        Monthly – Flexible
    </option>

    <option value="Quarterly">
        Quarterly – 3 Months
    </option>

    <option value="Yearly">
        Yearly – Best Value
    </option>

</select>

<span
    class="error"
    id="planError"
></span>

</div>

<!-- HEIGHT / WEIGHT CHECKBOXES -->

<div class="form-group full-width">

    <label>
        Check all that apply
    </label>

    <div class="checkbox-options">

        <label class="checkbox-option">

            <input
                type="checkbox"
                id="heightCheck"
            >

            Over 6 feet tall

        </label>

        <label class="checkbox-option">

            <input
                type="checkbox"
                id="weightCheck"
            >

            Over 200 pounds

        </label>

    </div>

</div>

<!-- ATHLETIC ABILITY -->

```

```

<div class="form-group full-width">

    <label for="ability">
        Describe your athletic ability
    </label>

    <textarea
        id="ability"
        placeholder="Tell us about your fitness experience, activities, or athletic abilities..."
    ></textarea>

    <span
        class="error"
        id="abilityError"
    ></span>

</div>

<!-- TERMS -->

<div class="form-group full-width">

    <div class="checkbox-option">

        <input
            type="checkbox"
            id="terms"
        >

        <label for="terms">
            I agree to the gym membership
            terms and conditions.
        </label>

    </div>

    <span
        class="error"
        id="termsError"
    ></span>

</div>

</div>

<!-- SUBMIT -->

<button type="submit">
    Enter My Information
</button>

<!-- MESSAGE -->

<div id="message"></div>

</form>

</div>

<script src="script2.js"></script>

</body>

```

</html>

script2.js

```
const form =
  document.getElementById("membershipForm");

const fullName =
  document.getElementById("fullName");

const age =
  document.getElementById("age");

const email =
  document.getElementById("email");

const phone =
  document.getElementById("phone");

const eyeColor =
  document.getElementById("eyeColor");

const goal =
  document.getElementById("goal");

const plan =
  document.getElementById("plan");

const ability =
  document.getElementById("ability");

const terms =
  document.getElementById("terms");

const heightCheck =
  document.getElementById("heightCheck");

const weightCheck =
  document.getElementById("weightCheck");

const message =
  document.getElementById("message");

// =====
// ERROR ELEMENTS
// =====

const nameError =
  document.getElementById("nameError");

const ageError =
  document.getElementById("ageError");

const genderError =
  document.getElementById("genderError");

const emailError =
  document.getElementById("emailError");

const phoneError =
  document.getElementById("phoneError");

const eyeError =
  document.getElementById("eyeError");
```

```

const goalError =
    document.getElementById("goalError");

const planError =
    document.getElementById("planError");

const abilityError =
    document.getElementById("abilityError");

const termsError =
    document.getElementById("termsError");

// =====
// VALIDATE NAME
// =====

function validateName() {

    const value =
        fullName.value.trim();

    if (value === "") {

        nameError.textContent =
            "Full name is required.";

        fullName.classList.add("invalid");
        fullName.classList.remove("valid");

        return false;
    }

    if (!/^[A-Za-z ]+$/ .test(value)) {

        nameError.textContent =
            "Name should contain only letters.";

        fullName.classList.add("invalid");
        fullName.classList.remove("valid");

        return false;
    }

    nameError.textContent = "";

    fullName.classList.add("valid");
    fullName.classList.remove("invalid");

    return true;
}

// =====
// VALIDATE AGE
// =====

function validateAge() {

    const value =
        Number(age.value);

    if (age.value === "") {

        ageError.textContent =

```

```

        "Age is required.";

        age.classList.add("invalid");
        age.classList.remove("valid");

        return false;
    }

    if (value < 16 || value > 60) {

        ageError.textContent =
            "Age must be between 16 and 60.";

        age.classList.add("invalid");
        age.classList.remove("valid");

        return false;
    }

    ageError.textContent = "";

    age.classList.add("valid");
    age.classList.remove("invalid");

    return true;
}

// =====
// VALIDATE GENDER
// =====

function validateGender() {

    const gender =
        document.querySelector(
            'input[name="gender"]:checked'
        );

    if (!gender) {

        genderError.textContent =
            "Please select your gender.";

        return false;
    }

    genderError.textContent = "";

    return true;
}

// =====
// VALIDATE EMAIL
// =====

function validateEmail() {

    const value =
        email.value.trim();

    const pattern =
        /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

```

```

    if (value === "") {

        emailError.textContent =
            "Email is required.";

        email.classList.add("invalid");
        email.classList.remove("valid");

        return false;
    }

    if (!pattern.test(value)) {

        emailError.textContent =
            "Enter a valid email address.";

        email.classList.add("invalid");
        email.classList.remove("valid");

        return false;
    }

    emailError.textContent = "";

    email.classList.add("valid");
    email.classList.remove("invalid");

    return true;
}

// =====
// VALIDATE PHONE
// =====

function validatePhone() {

    const value =
        phone.value.trim();

    if (value === "") {

        phoneError.textContent =
            "Phone number is required.";

        phone.classList.add("invalid");
        phone.classList.remove("valid");

        return false;
    }

    if (!/^[0-9]{10}$/.test(value)) {

        phoneError.textContent =
            "Phone number must contain 10 digits.";

        phone.classList.add("invalid");
        phone.classList.remove("valid");

        return false;
    }

    phoneError.textContent = "";

```

```

        phone.classList.add("valid");
        phone.classList.remove("invalid");

        return true;
    }

// =====
// VALIDATE EYE COLOR
// =====

function validateEyeColor() {

    if (eyeColor.value === "") {

        eyeError.textContent =
            "Please select your eye color.";

        eyeColor.classList.add("invalid");
        eyeColor.classList.remove("valid");

        return false;
    }

    eyeError.textContent = "";

    eyeColor.classList.add("valid");
    eyeColor.classList.remove("invalid");

    return true;
}

// =====
// VALIDATE FITNESS GOAL
// =====

function validateGoal() {

    if (goal.value === "") {

        goalError.textContent =
            "Please select a fitness goal.";

        goal.classList.add("invalid");
        goal.classList.remove("valid");

        return false;
    }

    goalError.textContent = "";

    goal.classList.add("valid");
    goal.classList.remove("invalid");

    return true;
}

// =====
// VALIDATE MEMBERSHIP PLAN
// =====

function validatePlan() {

    if (plan.value === "") {

```

```

        planError.textContent =
            "Please select a membership plan.";

        plan.classList.add("invalid");
        plan.classList.remove("valid");

        return false;
    }

    planError.textContent = "";

    plan.classList.add("valid");
    plan.classList.remove("invalid");

    return true;
}

// =====
// VALIDATE ATHLETIC ABILITY
// =====

function validateAbility() {

    const value =
        ability.value.trim();

    if (value === "") {

        abilityError.textContent =
            "Please describe your athletic ability.";

        ability.classList.add("invalid");
        ability.classList.remove("valid");

        return false;
    }

    if (value.length < 10) {

        abilityError.textContent =
            "Please enter at least 10 characters.";

        ability.classList.add("invalid");
        ability.classList.remove("valid");

        return false;
    }

    abilityError.textContent = "";

    ability.classList.add("valid");
    ability.classList.remove("invalid");

    return true;
}

// =====
// VALIDATE TERMS
// =====

function validateTerms() {

```

```

    if (!terms.checked) {

        termsError.textContent =
            "You must agree to the terms and conditions.";

        return false;
    }

    termsError.textContent = "";

    return true;
}

// =====
// NAME EVENTS
// =====

fullName.addEventListener(
    "focus",
    function() {

        fullName.style.backgroundColor =
            "#f5f8f1";

    }
);

fullName.addEventListener(
    "input",
    validateName
);

fullName.addEventListener(
    "blur",
    function() {

        validateName();

        fullName.style.backgroundColor = "";

    }
);

// =====
// AGE EVENTS
// =====

age.addEventListener(
    "focus",
    function() {

        age.style.backgroundColor =
            "#f5f8f1";

    }
);

age.addEventListener(
    "input",
    validateAge
);

```

```

age.addEventListener(
    "blur",
    function() {

        validateAge();

        age.style.backgroundColor = "";

    }
);

// =====
// GENDER EVENT
// =====

document
    .querySelectorAll(
        'input[name="gender"]'
    )
    .forEach(function(radio) {

        radio.addEventListener(
            "change",
            validateGender
        );

    });

// =====
// EMAIL EVENTS
// =====

email.addEventListener(
    "focus",
    function() {

        email.style.backgroundColor =
            "#f5f8f1";

    }
);

email.addEventListener(
    "input",
    validateEmail
);

email.addEventListener(
    "blur",
    function() {

        validateEmail();

        email.style.backgroundColor = "";

    }
);

// =====
// PHONE EVENTS
// =====

phone.addEventListener(
    "focus",

```

```

function() {

    phone.style.backgroundColor =
        "#f5f8f1";

}

);

phone.addEventListener(
    "input",
    function() {

        phone.value =
            phone.value.replace(/\D/g, "");

        validatePhone();

    }

);

phone.addEventListener(
    "blur",
    function() {

        validatePhone();

        phone.style.backgroundColor = "";

    }

);

// =====
// EYE COLOR EVENTS
// =====

eyeColor.addEventListener(
    "change",
    validateEyeColor
);

eyeColor.addEventListener(
    "blur",
    validateEyeColor
);

// =====
// FITNESS GOAL EVENTS
// =====

goal.addEventListener(
    "change",
    validateGoal
);

goal.addEventListener(
    "blur",
    validateGoal
);

// =====
// MEMBERSHIP PLAN EVENTS
// =====

```

```

plan.addEventListener(
    "change",
    validatePlan
);

plan.addEventListener(
    "blur",
    validatePlan
);

// =====
// ATHLETIC ABILITY EVENTS
// =====

ability.addEventListener(
    "input",
    validateAbility
);

ability.addEventListener(
    "blur",
    validateAbility
);

ability.addEventListener(
    "focus",
    function() {

        ability.style.backgroundColor =
            "#f5f8f1";

    }
);

// =====
// TERMS EVENT
// =====

terms.addEventListener(
    "change",
    validateTerms
);

// =====
// EXTRA CHECKBOX EVENTS
// =====

heightCheck.addEventListener(
    "change",
    function() {

        if (heightCheck.checked) {

            heightCheck.parentElement.style.fontWeight =
                "600";

        } else {

            heightCheck.parentElement.style.fontWeight =
                "500";

        }

    }
);

```

```

    }
  );

  weightCheck.addEventListener(
    "change",
    function() {

      if (weightCheck.checked) {

        weightCheck.parentElement.style.fontWeight =
          "600";

      } else {

        weightCheck.parentElement.style.fontWeight =
          "500";

      }

    }
  );

```

```

// =====
// FORM SUBMIT
// =====

```

```

form.addEventListener(
  "submit",
  function(event) {

    event.preventDefault();

    const validName =
      validateName();

    const validAge =
      validateAge();

    const validGender =
      validateGender();

    const validEmail =
      validateEmail();

    const validPhone =
      validatePhone();

    const validEyeColor =
      validateEyeColor();

    const validGoal =
      validateGoal();

    const validPlan =
      validatePlan();

    const validAbility =
      validateAbility();

    const validTerms =
      validateTerms();

    if (
      validName &&
      validAge &&
      validGender &&

```

```

    validEmail &&
    validPhone &&
    validEyeColor &&
    validGoal &&
    validPlan &&
    validAbility &&
    validTerms
  ) {

    message.textContent =
      "Membership successfully registered!";

    message.className =
      "success";

    // Collect selected checkbox information

    let additionalInfo = [];

    if (heightCheck.checked) {

      additionalInfo.push(
        "Over 6 feet tall"
      );

    }

    if (weightCheck.checked) {

      additionalInfo.push(
        "Over 200 pounds"
      );

    }

    // Display successful submission

    console.log(
      "Name:",
      fullName.value
    );

    console.log(
      "Age:",
      age.value
    );

    console.log(
      "Gender:",
      document.querySelector(
        'input[name="gender"]:checked'
      ).value
    );

    console.log(
      "Eye Color:",
      eyeColor.value
    );

    console.log(
      "Fitness Goal:",
      goal.value
    );

    console.log(

```

```

        "Membership Plan:",
        plan.value
    );

    console.log(
        "Athletic Ability:",
        ability.value
    );

    console.log(
        "Additional Information:",
        additionalInfo
    );

    form.reset();

    document
        .querySelectorAll(
            "input, select, textarea"
        )
        .forEach(function(field) {

            field.classList.remove(
                "valid"
            );

            field.classList.remove(
                "invalid"
            );

        });

    }

    else {

        message.textContent =
            "Please correct the highlighted fields.";

        message.className =
            "fail";

    }

    }

    );

```

Additional Fields

Field	Purpose
Eye Color	Allows the user to select Green, Blue, Brown, Hazel, or Gray.
Check all that apply	Provides checkboxes for Over 6 feet tall and Over 200 pounds.
Describe your athletic ability	Allows the user to describe fitness experience, activities, or athletic abilities.
Terms and Conditions	Requires agreement to the gym membership terms and conditions.

JavaScript Concepts / Functions Used

Concept / Function	Purpose
DOM Manipulation	Accesses and updates form elements dynamically.

Event Handling	Responds to user actions such as form submission and input changes.
Validation	Checks whether required information is entered correctly.
Checkbox Handling	Processes physical characteristic selections and terms agreement.
Textarea Handling	Collects and validates the member's athletic ability description.
Error Messages	Displays feedback when input is invalid.
Success Message	Confirms successful submission when validation requirements are satisfied.

Output

- The enhanced FitSpace Gym form collects additional member information.
- Eye color can be selected from a dropdown list.
- Users can indicate physical characteristics using checkboxes.
- An athletic ability description can be entered in a textarea.
- The membership plan and fitness goal are selected from dropdown menus.
- The form requires agreement to the membership terms and conditions.

Screenshot

FITSPACE GYM
Build strength. Stay consistent. Become your best.

MEMBERSHIP REGISTRATION

Full Name
Ojaswini Thote

Age
21

Gender
☐ Male ☒ Female ☐ Other

Email Address
ojaswinithote@gmail.com

Phone Number
8767064228

Eye Color
Brown

Fitness Goal
Strength Training

Membership Plan
Quarterly - 3 Months

Check all that apply
☐ Over 6 feet tall
☐ Over 200 pounds

Describe your athletic ability
I have good overall fitness and regularly participate in running, strength training, and other physical activities. I have good stamina and am comfortable with moderate to intense workouts.

☒ I agree to the gym membership terms and conditions.

ENTER MY INFORMATION

Figure 2: FitSpace Gym Membership Registration — Case Study Output

Result / Conclusion

Both the Practical and Case Study were completed successfully. The Practical demonstrates the development of a responsive gym membership registration form with client-side validation and event handling. The Case Study extends the form with additional information including eye color, physical characteristics, and athletic ability. JavaScript is used to interact with form elements, validate user input, display error messages, and provide feedback to the user. Thus, the experiment successfully demonstrates the use of JavaScript for creating interactive, validated, and user-friendly web forms.